Solutions EE208A 2020

MATHEMATICS 2A

CV-1 (1)

Determine if the function is harmonic and find the harmonic conjugate. (Q1)

$u = xe^x\cos y - ye^x\sin y$

sol:

$u_x = e^x\cos y + xe^x\cos y - ye^x\sin y$

$\Rightarrow u_{xx} = e^x\cos y + e^x\cos y + xe^x\cos y - ye^x\sin y$

$= 2e^x\cos y + xe^x\cos y - ye^x\sin y$ //

$u_y = xe^x(-\sin y) - e^x\sin y - ye^x\cos y$

$\Rightarrow u_{yy} = -xe^x\cos y - e^x\cos y - e^x\cos y + ye^x\sin y$

$= -xe^x\cos y - 2e^x\cos y + ye^x\sin y$ //

For u to be harmonic, we need $u_{xx} + u_{yy} = 0$

$u_{xx} + u_{yy} = 2e^x\cos y + xe^x\cos y - ye^x\sin y$
$- xe^x\cos y - 2e^x\cos y + ye^x\sin y$

$= 0$ //

$\therefore$ u is harmonic. //

To find the harmonic conjugate, we apply the Cauchy Riemann equations:

Let harmonic conjugate be v. Then

$v_y = u_x$ — (1)

$v_x = -u_y$ — (2)

① : $V_y = e^x \cos y + xe^x \cos y - ye^x \sin y$

$\Rightarrow V = e^x \sin y + xe^x \sin y - e^x \int y \sin y \, dy$

$V = e^x \sin y + xe^x \sin y - e^x [-y\cos y + \sin y] + g(x)$ — (A)

② : $V_x = -U_y = xe^x \sin y + e^x \sin y + ye^x \cos y$

$\Rightarrow V = (xe^x - e^x)\sin y + ye^x \cos y + e^x \sin y + b(y)$

$V = xe^x \sin y + e^x y \cos y$ — (B)

(A) and (B) must be the same since they are referring to the same function.

So, $g(x) = b(y) =$ constant and.

the required harmonic conjugate is $V = xe^x \sin y + e^x y \cos y$

Q2

CV-3

Evaluate $\oint_C \frac{2+3\sin(\pi z)}{z(z-1)^2}\,dz$ where C

is a square with vertices $\pm 3 \pm 3i$.

Poles are $z=0$ (simple) & $z=1$ (double)

Both poles are inside the square.

$$\text{Res}(z=0) = \lim_{z\to 0} \frac{2+3\sin(\pi z)}{(z-1)^2} = 2$$

$$\text{Res}(z=1) = \lim_{z\to 1} \frac{d}{dz}\left[\frac{2+3\sin(\pi z)}{z}\right]$$

$$= \lim_{z\to 1} \frac{3\pi\cos(\pi z)\cdot z - (2+3\sin(\pi z))\cdot 1}{z^2}$$

$$= -3\pi - 2$$

So, $\oint_C \frac{2+3\sin(\pi z)}{z(z-1)^2}\,dz = 2\pi i\left[2 + (-3\pi - 2)\right]$

$$= -6\pi^2 i$$

Q3 Show that $\int_0^{2\pi} \frac{\cos(3\theta)}{5-4\cos\theta}\,d\theta = \frac{\pi}{12}$. Use complex var. techniques.

If $z = e^{j\theta}$, then express $\cos\theta$ in terms of z.

$$\cos(3\theta) = \frac{e^{j(3\theta)} + e^{-j(3\theta)}}{2} = \frac{z^3 + z^{-3}}{2} \quad \& \quad dz = je^{j\theta}\,d\theta$$
$$= jz\,d\theta$$
$$\Rightarrow \quad d\theta = \frac{dz}{jz}$$

So our integral now becomes

$$\oint_C \frac{z^3 + z^{-3}}{2\left[5 - \frac{4}{2}\left(z + \frac{1}{z}\right)\right]} \frac{dz}{jz}$$ where C is circle of radius 1.

$$\frac{1}{2j}\oint_C \frac{z^3 + z^{-3}}{5z - 2z^2 - 2}\,dz = \frac{-1}{2j}\oint_C \frac{z^6 + 1}{z^3(2z^2 - 5z + 2)}\,dz$$
$$= \frac{-1}{2j}\oint_C \frac{z^6 + 1}{z^3(2z-1)(z-2)}\,dz$$

Poles are $z = 0, \frac{1}{2}$ and 2. All simple except $z = 0$ (triple)

But only $z = 0, \frac{1}{2}$ are inside C.

$$\frac{-1}{2j}\oint_C \frac{z^6 + 1}{z^3(2z-1)(z-2)}\,dz = \frac{-1}{2j} \cdot 2\pi j\left[\text{Res}(z=0) + \text{Res}\left(z=\tfrac{1}{2}\right)\right]$$

$$\text{Res}(z=0) = \frac{1}{(3-1)!}\lim_{z\to 0}\left(\frac{z^6+1}{(2z-1)(z-2)}\right)''$$

$$= \frac{1}{2}\frac{(6\cdot 7)}{(-8)(-1)} = \frac{21}{8}$$

$$\text{Res}\left(z=\tfrac{1}{2}\right) = \lim_{z\to\frac{1}{2}}\frac{z^6+1}{2z^3(z-2)} = \frac{\left(\frac{1}{2}\right)^6+1}{2\left(\frac{1}{2}\right)^3\left(\frac{1}{2}-2\right)}$$

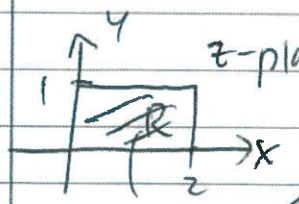
$$= \frac{\frac{1+2^6}{2^6}}{2\cdot\frac{1}{8}\cdot\left(-\frac{3}{2}\right)} = \frac{-65}{24}$$

So, $\displaystyle\int_0^{2\pi}\frac{\cos(3\theta)}{5-4\cos(\theta)}\,d\theta$

$$= \frac{-1}{2i}\oint_C \frac{z^6+1}{-z^3(5z-2z^2-2)}\,dz$$

$$= -\pi\left(\frac{21}{8}-\frac{65}{24}\right) = -\frac{\pi}{24}(63-65) = \frac{\pi}{12}$$

Q4 Let $w = \sqrt{2}\, e^{j(\frac{\pi}{4})} z + (1-2j)$. Let the region R be bounded by $x=0$, $y=0$, $x=2$ & $y=1$. Determine the region R' of the w-plane into which R is mapped under the transformation given by w.



R //

$$w = u + jv = (1+j)(x+jy) + (1-2j)$$

$$= (x-y+1) + j(x+y-2)$$

$$\Rightarrow u = x-y+1, \quad v = x+y-2$$

So, $x=0 \rightarrow u = 0-y+1, \; v = y-2$

$$\Rightarrow 1-u = v+2 \Rightarrow u+v = -1$$

$y=0 \rightarrow u = x+1, \; v = x-2$

$$\Rightarrow u-1 = v+2 \Rightarrow u-v = 3$$

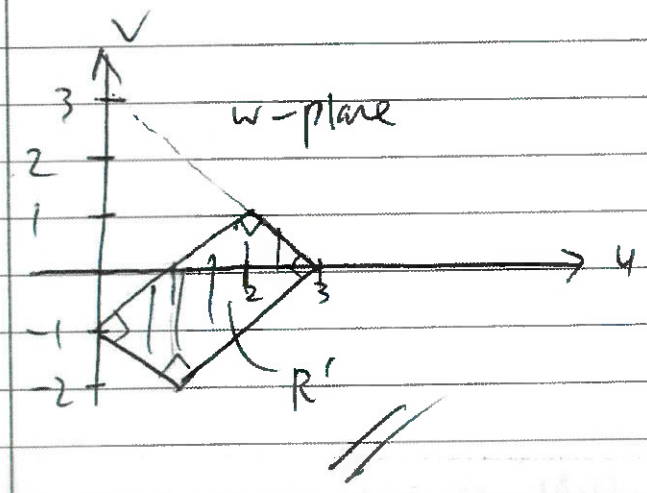
$x=2 \rightarrow u = 2-y+1, \; v = 2+y-2$

$$\Rightarrow 2-u+1 = v \Rightarrow u+v = 3$$

$y=1 \rightarrow u = x-1+1, \; v = x+1-2$

$$\Rightarrow u = v+1 \Rightarrow u-v = 1$$

(2)

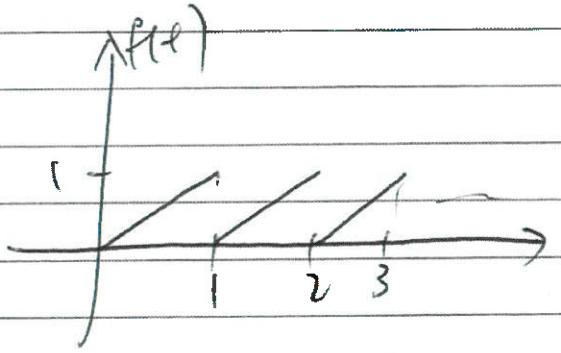


So we see that the mapping has rotation, stretching & translation. (BONUS)

(1)

LT-1 & 2

(Q5)



$$f(t)=\begin{cases} t, & 0\le t<1 \\ f(t-1), & t\ge 1 \end{cases}$$

Find the LT.

sol: $T=\text{period}=1$

$$\text{So } LT=\frac{1}{1-e^{-s}}\left[\int_0^1 te^{-st}dt\right]=\frac{1-e^{-s}(1+s)}{s^2(1-e^{-s})}$$

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Q6 [LT-3] (3)

3. Use Laplace Transforms to solve the ODE

$$y'' + y = t, \qquad y(0) = 0, \; y'(0) = 2$$

Sol:

$$s^2\bar{Y} - sy(0) - y'(0) + \bar{Y} = \frac{1}{s^2}$$

$$\Rightarrow (s^2+1)\bar{Y}(s) = \frac{1}{s^2} + 2$$

$$\Rightarrow \bar{Y}(s) = \frac{1}{s^2(s^2+1)} + \frac{2}{s^2+1}$$

Use partial fraction for $\frac{1}{s^2(s^2+1)}$.

$$\frac{1}{s^2(s^2+1)} = \frac{As+B}{s^2} + \frac{Cs+D}{s^2+1}$$

$$\Rightarrow 1 = (As+B)(s^2+1) + (Cs+D)(s^2)$$

$$\Rightarrow 1 = s^3(A+C) + s^2(B+D) + s(A) + B$$

$$\Rightarrow B = 1, \; A = C = 0 \; \& \; D = -1$$

So, $\frac{1}{s^2(s^2+1)} + \frac{2}{s^2+1} = \frac{1}{s^2} - \frac{1}{s^2+1} + \frac{2}{s^2+1} = \frac{1}{s^2} + \frac{1}{s^2+1}$

$$\Rightarrow \bar{y}(s) = \frac{1}{s^2} + \frac{1}{s^2+1} \rightarrow \underline{y(t) = t + \sin t}$$

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